

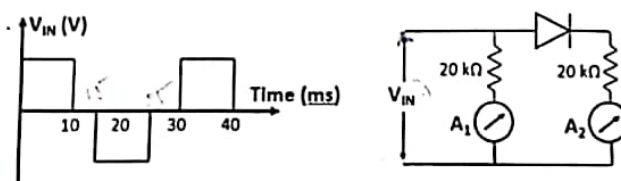
ELECTRONICS ENGINEERING DEPARTMENT IIT (ISM) DHANBAD
SEMESTER MONSSON 2018-1019

EXAMINATION: END SEMESTER EXAMINATION
SEMESTER B.TECH. 1ST SEMESTER (GROUP E, F, G, H)
SUBJECT: ECC 11101 ELECTRONIC ENGINEERING

TIME: 3 HRS
MAX MARKS: 100

INSTRUCTIONS: ANSWER ALL QUESTIONS FROM PART I, AND ANY TWO QUESTIONS FROM PART II.
MAKE SUITABLE ASSUMPTIONS WHEREVER NECESSARY

PART I

Q.1	State whether the following statements are TRUE or FALSE (T/F). Give appropriate reasoning (R) for your answer in brief. a. An ideal op amp has voltage gain of the order of 10^6 A/A. b. Waveforms can be shaped easily by using a combination of diode, capacitor and battery. c. A rectifier converts alternating current into a unidirectional current. In a half-wave rectification with input voltage $V_m \sin \omega t$ and a resistive load R_L , the dc current $I_{dc} = V_m / \pi R_L$. d. In a digital circuit we have the output given as $\overline{(\overline{A} + \overline{B})} + \overline{C}$. This expression is equivalent to $BC + AC$. e. The number $(5F)_{16}$ is equivalent to $(96)_{10}$.	$5 \times (1(T/F) + 3(R))$ = 20 Marks
Q.2 (a)	The signal V_{IN} is applied to the circuit shown in Figure 1. Predict the reading on the two meters A_1 and A_2 , if they are DC ammeters.  Figure 1	10 marks
Q.2 (b)	A four-variable logic function comprises of following max-terms: $\prod M(3,7,9,13,15)$. Deduce the minimized SOP form using k-map. Draw the corresponding logic circuit. $(A+B)(A+C)(A+D)$	10 marks
Q3 (a)	A Silicon semiconductor doped with 10^{17} cm^{-3} phosphorus atoms has been used to design a humidity sensor. In May 2018, the sensor was tested at Delhi, India (35°C); Aziziyah, Libya (50°C); Melbourne, Australia (14°C) and Talvik, Norway (-5°C). Numbers inside the parenthesis represent the temperature of the place. When an electric field of 1 MV/m was applied, 16 kA/cm^2 current density was observed in the Silicon sensor. Assuming the diffusivity to be independent of temperature, in which place/city the measurement was performed? Electron mobility $0.135 \text{ m}^2/\text{V.s}$, Hole mobility $0.048 \text{ m}^2/\text{V.s}$	10 marks

$$D_n = 218 \text{ cm}^2/\text{s}, D_p = 1 \text{ cm}^2/\text{s}$$

$$16 \times 10^3 \text{ A/m}^2 = e n D_n \frac{dV}{dx} + e p D_p \frac{dV}{dx}$$

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